

Data Statement

The dataset used in this study is a combination of two official NBA sources: the Shot Detail dataset and the Play-by-Play (cdnnba) dataset.

Part 1:2024 Shot Detail Dataset

This dataset mainly comes from the shot detail API (Shot Chart Detail) provided by stats.nba.com. It records each player's shot attempt, including shot action type (e.g., Jump Shot, Layup), shot zone (e.g., Restricted Area, Mid-Range, 3-Point), actual shooting distance (SHOT_DISTANCE , in feet), court coordinates (LOC_X, LOC_Y), shot result (made or missed), and total points (pointsTotal). In addition, it includes some assist information (assistPlayerNameInitial , assistPersonId , assistTotal). Part of the data provides detailed spatial and action features for single shots, supporting further analysis of shooting tendencies, hot zones, and spatial modeling.

Part 2: Play-by-Play Dataset (cdnnba_2024)

This dataset comes from data.nba.com and pbpstats.com, which record every in-game event in sequence (Play-by-Play). It contains detailed information about each event, such as scoring, fouls, turnovers, assists, and rebounds. In particular, it includes the home and away team scores(scoreHome, scoreAway), from which the net score margin (SCORE_MARGIN) can be derivea. Based on this, the “score margin from the shooter's perspective" variable (MARGIN_ACTION) is constructed. These score and time variables are essential for analyzing whether players take shots while leading or trailing. This dataset connects shot events with the overall game context (time and score), supporting contextual analysis of shooting behavior.

1.1 Dataset Merging

We merged the two datasets using "same game + same event" as the precise join key. The method is left join, ensuring that all shots are kept even if the right table lacks certain fields. After merging, we compared TEAM_ID with teamId in the right table and found all data matched with no missing or inconsistent records. For PLAYER_ID vS. personId, we listed the different rows for manual review.

1.2 Removing Duplicate and Redundant Columns

To reduce redundancy, multicollinearity, and information leakage risk, we removed variables irrelevant to the study or duplicated across the two datasets, keeping only features relevant to the research questions.

1.2.1 Duplicate Columns

We deleted duplicate variables in both datasets, such as actionNumber, orderNumber, gameId, period, personId, teamId, teamTricode, playerName, playerNameI, personIdsFilter, area, areaDetail, shotDistance. These columns with GAME_ID, GAME_EVENT_ID, PLAYER_ID, TEAM_ID, TEAM_NAME. To avoid unnecessary size and confusion, these duplicates were removed.

1.2.2 Text Descriptions and Metadata

We removed description, subType, qualifiers, descriptor, edited, isTargetScoreLastPeriod, timeActual columns. These are mainly descriptive text, which can be derived from structured variables. Removing them prevents noise and unnecessary effects on modeling.

1.2.3 Jump Ball Fields

We removed jumpBallRecoveredName, jumpBallRecoverdPersonId, jumpBallWonPlayerName, jumpBallWonPersonId, jumpBallLostPlayerName, jumpBallLostPersonId columns. Jump ball events occur very rarely and have no contribution to the research question.

1.2.4 Defensive Action Names and IDs

We removed DlockPlayerName, blockPersonId, stealPlayerName, stea LPersonId columns. These only record opponent defensive actions, which are not the focus of this study. Due to high missing rates and lack of standardization, they were removed.

1.2.5 Foul and Drawn-Foul Related

We removed foulPersonalTotal, foulTechnicalTotal, foulDrawnPlayerName, foulDrawnPersonId columns. These have many missing values and are not central to single-shot analysis.

1.3 Data Type Check

After merging and removing duplicates and redundant columns, the dataset contains the following variables:

1.3.1 Game and Identification Information

GRID_TYPE: Source type. Always “Shot Chart Detail”. Limited analytical value but confirms origin.

GAME_ID: Unique game identifier. Shared across events in the same game.

GAME_EVENT_ID: Unique event ID within a game. Increments from 1.

GAME_DATE: Game date. Converted to standard format (YYYY-MM-DD).

HTM, VTM: Home and visiting team codes.

TEAM_ID, TEAM_NAME: Team ID and name.

PLAYER_ID, PLAYER_NAME: Player ID and name. IDs used for modeling, names for presentation.

PERIOD, periodType: Game period. Integers 1–4 for regular time, 5+ for overtime. periodType gives text format.

clock: Remaining time in the period, converted to seconds.

1.3.2 Shot Actions and Behavior

ACTION_TYPE: Specific shot action (e.g., Jump Shot, Driving Layup).

SHOT_TYPE: “2PT Field Goal” or “3PT Field Goal”.

SHOT_ZONE_BASIC: Basic zone categories.

SHOT_ZONE_AREA: Finer directional zones.

SHOT_ZONE_RANGE: Distance ranges. Related to SHOT_DISTANCE.

SHOT_DISTANCE: Actual shot distance in feet. Outliers >40 feet represent half-court or buzzer shots.

shotResult: Shot made or missed.

pointsTotal: Points scored (0, 2, or 3).

side: Half-court side (left or right).

Assist fields: assistPlayerNameInitial, assistPersonId, assistTotal.

1.3.3 Score and Context Variables

scoreHome, scoreAway: Home and away scores.

SCORE_MARGIN: Home score minus away score.

MARGIN_ACTION: Score margin from the shooter’s perspective. Positive = shooter’s team leading, negative equal to trailing.

1.3.4 Coordinates

LOC_X, LOC_Y: Court coordinates.

x, y: Normalized coordinates (0–100).

1.4 Missing Values

After merging, only 3 shot records had major missing values. These retained basic shot attributes but lacked game context (scores, margin, clock, coordinates). These events were extreme cases (e.g., backcourt heaves) recorded in Shot Detail but not matched in Play-by-Play.

Filling rejected: statistical imputation would create bias.

Manual tracing infeasible: requires video review, outside project scope.

Decision: delete these 3 records (<0.01%).

Other blanks, such as pointsTotal for missed shots or assist fields for unassisted shots, are structural nulls. We replaced them with 0 to indicate no points or no assist.

1.5 Outliers

We found extreme but valid cases: SHOT_DISTANCE: Up to 80 feet, representing half-court or buzzer-beater attempts.

LOC_Y: Range –52 to 842, beyond the typical 0–470 court limit. These reflect long-distance attempts.

These are not errors but rare real events. We kept them, marking as outliers. For visualization, Y-axis limits (e.g., –50 to 470) improve readability.

1.6 Collinearity

Some variables show high correlation, such as SHOT_DISTANCE and SHOT_ZONE_RANGE. These will be considered in later modeling.

References

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